

SECR2043 OPERATING SYSTEMS

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 Section : 02

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write "**No output**" instead.

1.	Display your current working directory.	
	Command	Output
	<code>pwd</code>	<pre>chau@secr2043:~\$ pwd /home/chau</pre>
2.	List all the files and directories in your home directory, including hidden ones.	
	Command	Output
	<code>ls -R (all files in the subdirectories home)</code> <code>ls -a (hidden files)</code>	<pre>chau@secr2043:~\$ ls -R .: example.txt share ./share: textfile01.txt chau@secr2043:~\$ ls -abash_logout .bashrc .cache .profile .ssh example.txt share</pre>
3.	Create a new directory named "os_lab" in your home directory.	
	Command	Output
	<code>Mkdir os_lab</code> <code>ls</code>	<pre>combined.txt file2.txt filelist.txt os_lab workspace example.txt file3.txt linux-lab share</pre>
4.	Change your current working directory to "os_lab".	
	Command	Output
	<code>cd os_lab</code> <code>pwd</code>	<pre>/home/chau/os_lab</pre>
5.	Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.	
	Command	Output

	touch hello.txt echo "Hello, world" > hello.txt	No output
6.	Display the content of "hello.txt".	
	Command	Output
	cat hello.txt	Hello,world
7.	Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.	
	Command	Output
	cp hello.txt hello_copy.txt	no output
8.	Rename "hello_copy.txt" to "hello_bak.txt".	
	Command	Output
	mv hello_copy.txt hello_bak.txt	No output
9.	Delete "hello.txt".	
	Command	Output
	rm hello.txt	No output
10.	Display your home directory with additional information, together with subdirectories contents.	
	Command	Output

<pre>ls -alhR ~</pre>	<pre> chau@secl2043:~/os_lab\$ ls -alhR ~ /home/chau: total 60K drwxr-x--- 8 chau chau 4.0K Apr 22 07:04 . drwxr-xr-x 6 root root 4.0K Apr 16 06:16 .. -rw-r--r-- 1 chau chau 220 Apr 16 06:16 .bash_logout -rw-r--r-- 1 chau chau 3.7K Apr 16 06:16 .bashrc drwx----- 2 chau chau 4.0K Apr 16 07:52 .cache -rw-r--r-- 1 chau chau 807 Apr 16 06:16 .profile drwx----- 2 chau chau 4.0K Apr 16 07:53 .ssh -rw-r--r-- 1 chau chau 158 Apr 22 02:29 combined.txt -rw-r--r-- 1 chau chau 0 Apr 22 00:21 example.txt -rw-r--r-- 1 chau chau 79 Apr 22 02:25 file2.txt -rw-r--r-- 1 chau chau 79 Apr 22 02:28 file3.txt -rw-r--r-- 1 chau chau 316 Apr 22 02:26 filelist.txt drwxr-xr-x 2 chau chau 4.0K Apr 22 02:22 linux-lab drwxr-xr-x 2 chau chau 4.0K Apr 22 07:10 os_lab drwxr-xr-x 2 chau chau 4.0K Apr 22 00:21 share drwxr-xr-x 2 chau chau 4.0K Apr 22 02:20 workspace /home/chau/.cache: total 8.0K drwx----- 2 chau chau 4.0K Apr 16 07:52 . drwxr-x--- 8 chau chau 4.0K Apr 22 07:04 .. -rw-r--r-- 1 chau chau 0 Apr 16 07:52 motd.legal-displayed /home/chau/.ssh: total 16K drwx----- 2 chau chau 4.0K Apr 16 07:53 . drwxr-x--- 8 chau chau 4.0K Apr 22 07:04 .. -rw----- 1 chau chau 978 Apr 16 07:53 known_hosts -rw-r--r-- 1 chau chau 142 Apr 16 07:53 known_hosts.old /home/chau/linux-lab: total 8.0K drwxr-xr-x 2 chau chau 4.0K Apr 22 02:22 . drwxr-x--- 8 chau chau 4.0K Apr 22 07:04 .. /home/chau/os_lab: total 12K drwxr-xr-x 2 chau chau 4.0K Apr 22 07:10 . drwxr-x--- 8 chau chau 4.0K Apr 22 07:04 .. -rw-r--r-- 1 chau chau 13 Apr 22 07:10 hallo_bak.txt /home/chau/share: total 8.0K drwxr-xr-x 2 chau chau 4.0K Apr 22 00:21 . drwxr-x--- 8 chau chau 4.0K Apr 22 07:04 .. -rw-r--r-- 1 chau chau 0 Apr 22 00:21 textfile01.txt /home/chau/workspace: total 8.0K drwxr-xr-x 2 chau chau 4.0K Apr 22 02:20 . drwxr-x--- 8 chau chau 4.0K Apr 22 07:04 .. </pre>
Total Mark:	

ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

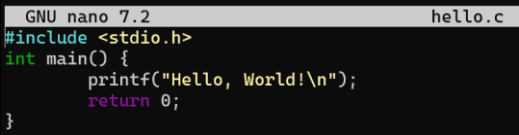
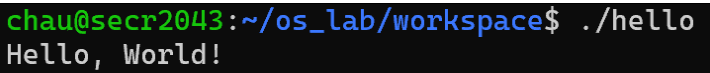
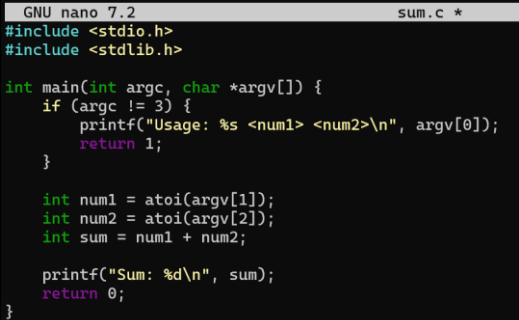
For each question, write down the command and the output (if any) in the answer sheet.

1.	Create an empty file named "hello.txt" in the current working directory.	
	Command	Output
	<code>touch hello.txt</code>	No output
2.	Write the text "Hello World" to "hello.txt" using echo.	
	Command	Output
	<code>echo "Hello, world" > hello.txt</code>	No output
3.	Append the text "This is a test" to "hello.txt" using echo.	
	Command	Output
	<code>echo "This is a test" >> hello.txt</code>	No output
4.	Display the contents of "hello.txt" using cat.	
	Command	Output
	<code>cat hello.txt</code>	<pre>chau@secr2043:~/os_lab\$ echo "Hello, world" > hello.txt chau@secr2043:~/os_lab\$ echo "This is a test" >> hello.txt chau@secr2043:~/os_lab\$ cat hello.txt Hello, world This is a test</pre>
5.	Create another file named "bye.txt" with the text "Goodbye World" using echo.	
	Command	Output
	<code>echo "Goodbye World" > bye.txt</code>	No output
6.	Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.	
	Command	Output
	<code>cat hello.txt bye.txt</code>	<pre>chau@secr2043:~/os_lab\$ echo "Goodbye World" > bye.txt chau@secr2043:~/os_lab\$ cat hello.txt bye.txt Hello, world This is a test Goodbye World</pre>
7.	Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.	
	Command	Output

	cat hello.txt bye.txt > greeting.txt	No output
8.	Display the contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	<pre> chau@secre2043:~/os_lab\$ cat hello.txt bye.txt > greeting.txt chau@secre2043:~/os_lab\$ cat greeting.txt Hello, world This is a test Goodbye World </pre>
9.	Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.	
	Command	Output
	nano greeting.txt	No output
	<pre> GNU nano 7.2 greeting.txt * Hello, Chau This is a test Goodbye World </pre>	
10.	Display the modified contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	<pre> chau@secre2043:~/os_lab\$ cat greeting.txt Hello, Chau This is a test Goodbye World </pre>
	Total Mark:	

ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc. For each question, write down the command and the output (if any) in the answer sheet.

1.	Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".	
	Command	Output
	<code>nano hello.c</code> 	No output
2.	Compile the program "hello.c" using gcc and generate an executable file named "hello".	
	Command	Output
	<code>gcc hello.c -o hello</code>	No output
3.	Run the executable file "hello" and display its output.	
	Command	Output
	<code>./hello</code>	
4.	Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".	
	Command	Output
	<code>nano sum.c</code> 	No output
5.	Compile the program "sum.c" using gcc and generate an executable file named "sum".	
	Command	Output
	<code>gcc sum.c -o sum</code>	No output
6.	Run the executable file "sum" with 10 and 20 as arguments and display its output.	
	Command	Output
	<code>./sum 10 20</code>	

7.	Run the executable file "sum" with -5 and 15 as arguments and display its output.	
	Command	Output
	<code>./sum -5 15</code>	<code>chau@secr2043:~/os_lab/workspace\$./sum -5 15 Sum: 10</code>
8.	Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".	
	Command	Output
	<code>nano echo.c</code> <pre> GNU nano 7.2 echo.c * #include <stdio.h> int main() { char line[256]; printf("Enter a line of text: "); fgets(line, sizeof(line), stdin); printf("You entered: %s", line); return 0; } </pre>	No output
9.	Compile the program "echo.c" using gcc and generate an executable file named "echo".	
	Command	Output
	<code>gcc echo.c -o echo</code>	No output
10.	Run the executable file "echo", type "Hello World" as input and display its output.	
	Command	Output
	<code>./echo</code> Hello World	<code>chau@secr2043:~/os_lab/workspace\$./echo Enter a line of text: Hello World You entered: Hello World</code>
	Total Mark:	